

NORTH SYDNEY GIRLS HIGH SCHOOL



2016 TRIAL HSC EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black pen.
- Board approved calculators may be used.
- A reference sheet is provided.
- Show all necessary working in questions 11–14.

Total Marks –70

Section I

Pages 3 - 6

10 Marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II

Pages 7–15

60 marks

- Attempt Questions 11–14.
- Allow about 1 hour and 45 minutes for this section.

NAME: _____

STUDENT NUMBER: _____

Colour in the circle next to your teacher's name:

- ☐ Mr T
- ☐ Ms Narayanan
- ☐ Ms Everingham
- ☐ Mrs Juhn
- ☐ Mr Moon
- ☐ Ms Viswanathan

QUESTION	MARK
1 – 10	/10
11	/15
12	/15
13	/15
14	/15
TOTAL	/70

Section I

10 marks

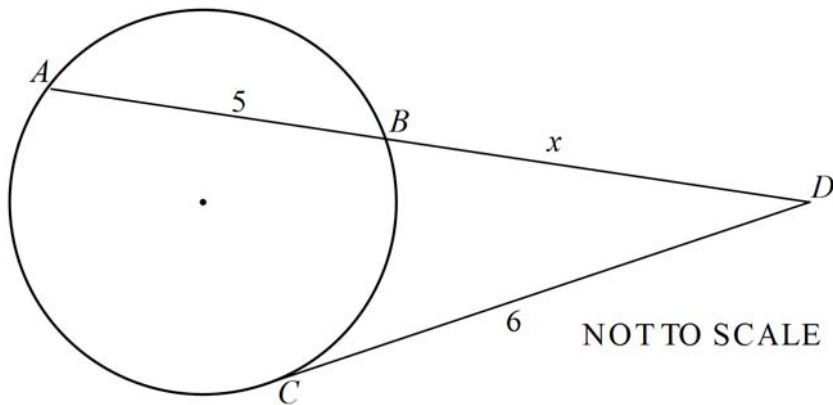
Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 Which of the following is equivalent to $\int \frac{1}{\sqrt{1-4x^2}} dx$?
- (A) $\sin^{-1}\left(\frac{x}{2}\right) + C$
- (B) $\sin^{-1}(2x) + C$
- (C) $\frac{1}{2}\sin^{-1}\left(\frac{x}{2}\right) + C$
- (D) $\frac{1}{2}\sin^{-1}(2x) + C$
- 2 The point P divides the interval from $A(3, -1)$ to $B(9, 2)$ externally in the ratio 5 : 2.
What is the x coordinate of P ?
- (A) $-7\frac{4}{5}$
- (B) $4\frac{5}{7}$
- (C) $7\frac{2}{7}$
- (D) 13
- 3 What is the value of $\lim_{x \rightarrow 0} \frac{x \cos 3x}{\sin 2x}$?
- (A) $\frac{1}{2}$
- (B) $\frac{2}{3}$
- (C) $\frac{3}{2}$
- (D) 2

- 4 In the diagram below, CD is a tangent to the circle at C and AB is 5 cm long. What is the length of BD in centimetres (marked with an x on the diagram)?



- (A) 1
- (B) $1\frac{1}{5}$
- (C) $2\frac{1}{5}$
- (D) 4
- 5 The velocity, v , of a particle in simple harmonic motion is given by $v^2 = 4 + 4x - 2x^2$. What is the amplitude and centre of motion?
- (A) amplitude is 3 and centre is $x = 1$
- (B) amplitude is 3 and centre is $x = -1$
- (C) amplitude is $\sqrt{3}$ and centre is $x = 1$
- (D) amplitude is $\sqrt{3}$ and centre is $x = -1$

- 6 Which of the following pairs represents the domain and range of $y = \ln(1 + \sqrt{4 - x^2})$?
- (A) Domain: $-2 \leq x \leq 2$
Range: $y \geq 0$
- (B) Domain: $-\sqrt{2} \leq x \leq \sqrt{2}$
Range: $y \geq \ln 3$
- (C) Domain: $-2 \leq x \leq 2$
Range: $0 \leq y \leq \ln 3$
- (D) Domain: $-\sqrt{2} \leq x \leq \sqrt{2}$
Range: All real y
- 7 Let α , β and γ be the roots of $P(x) = 2x^3 - 5x^2 + 4x - 9$. What is the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$?
- (A) $-\frac{5}{9}$
- (B) $-\frac{4}{9}$
- (C) $\frac{4}{9}$
- (D) $\frac{5}{9}$
- 8 What is the derivative of $y = \cos^{-1}\left(\frac{1}{x}\right)$?
- (A) $\frac{-1}{\sqrt{x^2 - 1}}$
- (B) $\frac{-1}{|x|\sqrt{x^2 - 1}}$
- (C) $\frac{1}{\sqrt{x^2 - 1}}$
- (D) $\frac{1}{|x|\sqrt{x^2 - 1}}$

9 Which of the following are true for all real values of x ?

I $\sin\left(\frac{\pi}{2} + x\right) = \cos\left(\frac{\pi}{2} - x\right)$

II $\sin\left(x + \frac{3\pi}{2}\right) = \cos(\pi - x)$

III $\sin x \cos x \leq \frac{1}{4}$

IV $2 + 2 \sin x - \cos^2 x \geq 0$

- (A) **I** and **II**
(B) **III** and **IV**
(C) **II** and **IV**
(D) **I** and **III**

10 Oil is spilled from an oil rig in the Gulf of Mexico and spreads in a circle with the circumference changing at a rate of 40 m/s. How fast is the area of the spill increasing when the circumference of the circle is 100π m?

- (A) $1500 \text{ m}^2 / \text{s}$
(B) $2000 \text{ m}^2 / \text{s}$
(C) $2100 \text{ m}^2 / \text{s}$
(D) $2500 \text{ m}^2 / \text{s}$

Section II

60 marks

Attempt Questions 11 – 14.

Allow about 1 hour and 45 minutes for this section.

Answer each question in a NEW writing booklet. Extra pages are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Solve $2x - 1 \geq \frac{6}{x}$. 3

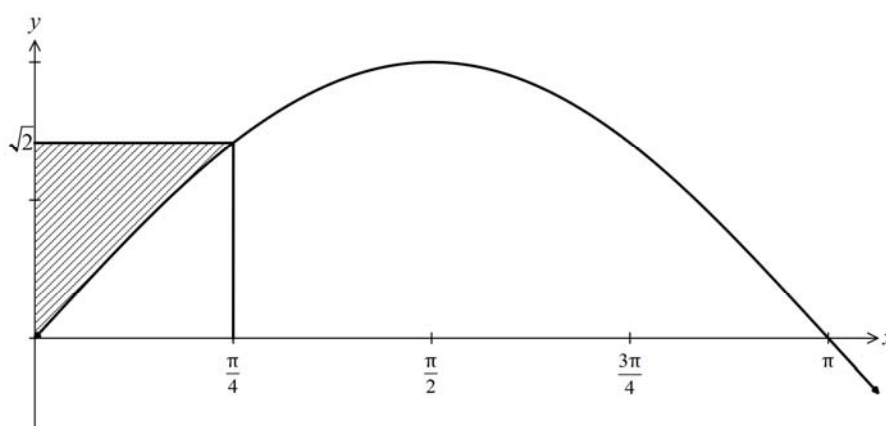
(b) (i) Show that $\sqrt{3} \sin x + \cos x$ can be written in the form $2 \cos\left(x - \frac{\pi}{3}\right)$. 3

(ii) Hence, or otherwise, find the value(s) of x for which $\sqrt{3} \sin x + \cos x$ 2

is a minimum in the interval $0 \leq x \leq 2\pi$.

(c) Use the substitution $u = \sqrt{x}$ to evaluate $\int_1^9 \frac{1}{x + \sqrt{x}} dx$. 3

(d) The diagram below shows the shaded region bounded by the curve $y = 2 \sin x$ and the y axis for $0 \leq y \leq \sqrt{2}$.



The region is rotated about the x axis to generate a solid of revolution.

(i) Show that the volume, V , of the solid is given by $V = \frac{\pi^2}{2} - \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x dx$. 1

(ii) Find the exact value of V . 3

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Factorise $P(x) = 2x^3 + 3x^2 - 1$, given that $x = -1$ is a zero. **2**

- (ii) Solve the equation $\sqrt{x^2 + 2x + 5} = x + \sqrt{2x + 3}$. **3**

- (b) Bobby has a can of cola at a temperature of 23°C . He places the can in a fridge which has a temperature of 3°C .

After t minutes, the temperature, c (in $^\circ\text{C}$), of the can of cola satisfies:

$$\frac{dc}{dt} = -\frac{1}{25}(c - 3)$$

- (i) Show that $c = 3 + ae^{-\frac{t}{25}}$ satisfies this equation, where a is a constant. **1**

- (ii) Bobby would like to drink the can of cola when its temperature is 5°C . **3**

If he put the can in the fridge at 8:50 a.m, when should he drink it?

Give your answer to the nearest minute.

- (c) (i) Prove by mathematical induction that: **3**

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2 \text{ for integers } n \geq 1$$

- (ii) Deduce that $2^3 + 4^3 + 6^3 + \dots + (2n)^3 = 2n^2(n+1)^2$. **1**

- (iii) Hence, or otherwise, find a simplified expression for: **2**

$$1^3 + 3^3 + 5^3 + \dots + (2n-1)^3$$

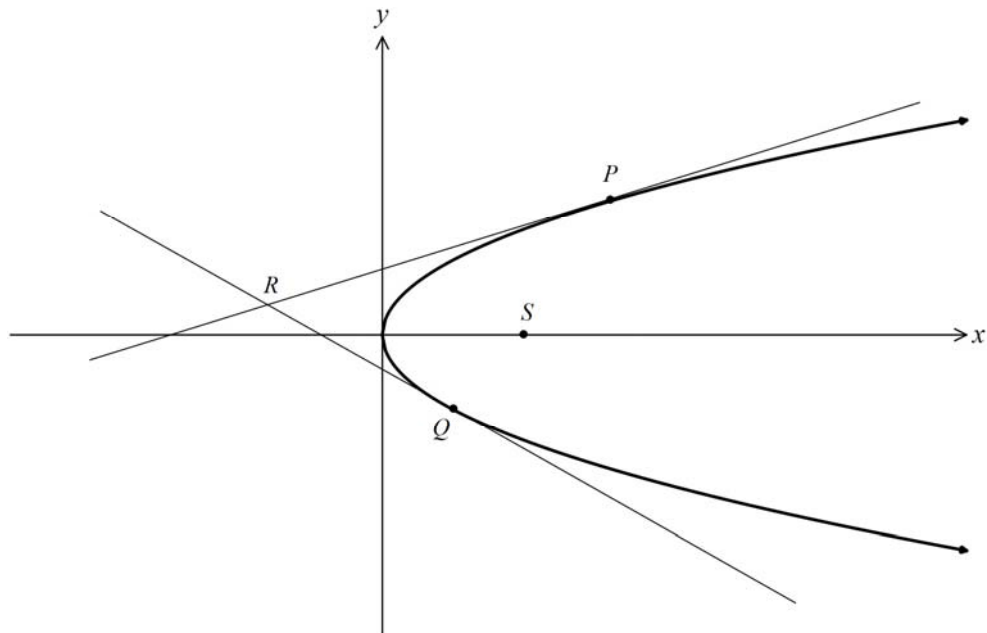
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The point $P(3p^2, 6p)$ lies on the parabola $y^2 = 12x$ and the point S is the focus of this parabola. The point $Q(3q^2, 6q)$, where $p \neq q$, also lies on the parabola.

The tangents to the parabola at the points P and Q meet at the point R , as shown in the diagram below.

The equation of the tangent at point P is given by $y = \frac{1}{p}x + 3p$.

(Do NOT prove this).

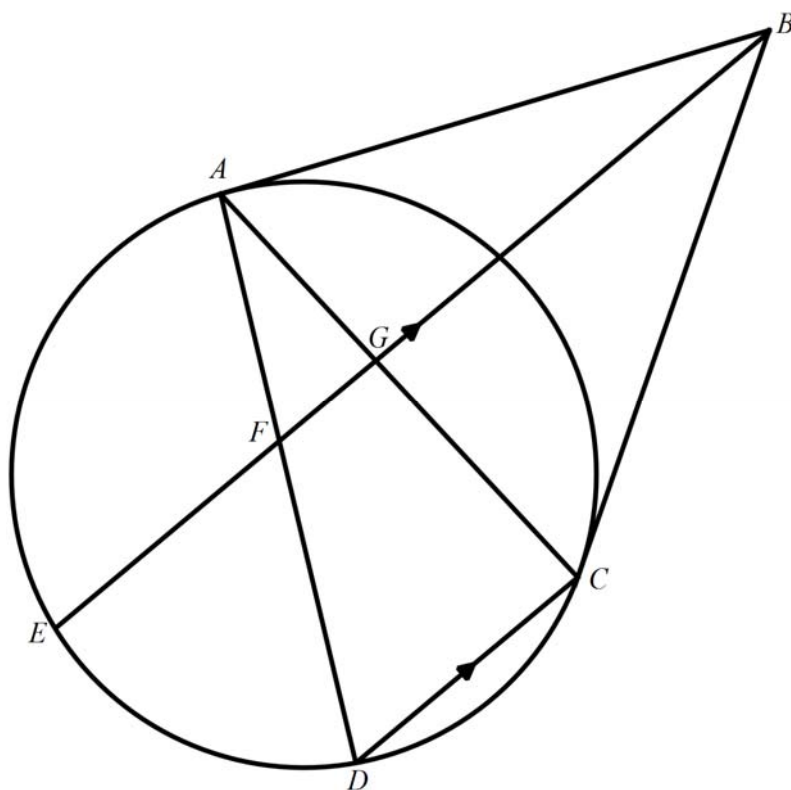


- | | | |
|-------|---|---|
| (i) | Prove that $SP = 3(1 + p^2)$. | 1 |
| (ii) | Find the coordinates of R . | 2 |
| (iii) | Hence, or otherwise, prove that $SR^2 = SP \times SQ$. | 2 |

Question 13 continues on page 11

Question 13 (continued)

- (b) In the diagram, EB is parallel to DC . Tangents from B meet the circle at A and C .



Let $\angle BCA = \alpha$.

Prove that:

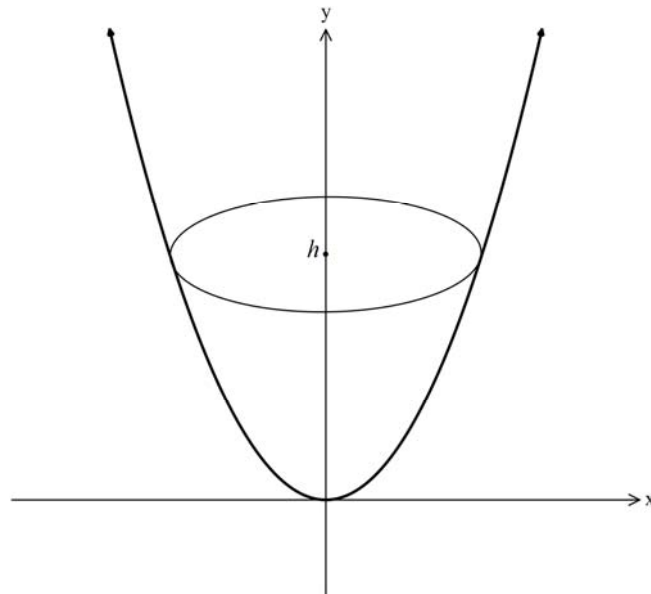
- | | | |
|------|-----------------------------------|---|
| (i) | $\angle BCA = \angle BFA$. | 2 |
| (ii) | $ABCF$ is a cyclic quadrilateral. | 1 |
- (c) Find a general solution of the following equation: 2

$$\sin\left(2x + \frac{\pi}{4}\right) = \frac{\sqrt{3}}{2}$$

Question 13 continues on page 12

Question 13 (continued)

- (d) A bowl is formed by rotating $y = x^{2n}$ (where n is an integer $n \geq 1$) about the y axis as shown.



The bowl is initially filled with water. The bowl has a small hole in its bottom (initially plugged).

When the hole is unplugged, water flows at a rate described by:

$$\frac{dV}{dt} = -\pi a^2 \sqrt{2gh}$$

where a is the radius of the hole, h is the depth of water at any time and g is a constant.

- (i) Show that the volume, V , of the water in the bowl is given by: 2

$$V = \frac{n\pi}{n+1} \cdot h^{\frac{1+n}{n}}$$

- (ii) When the hole is unplugged, find the expression which describes the rate at which the water level is changing with respect to time. 2
- (iii) If the water level falls at a constant rate with respect to time, find the value of n . 1

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider a function $f(x)$. The domain of $y = f(x)$ is $a < x < b$ and the range is all real y . Assume that $f'(x)$ exists for $a < x < b$ and is non-zero.

Let $g(x)$ be the inverse function of $f(x)$. Assume that $g'(x)$ exists.

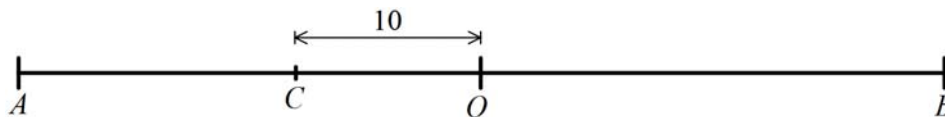
(i) Show that $g'(x) = \frac{1}{f'(g(x))}$. 2

(ii) Consider $f(x) = 3 + x^2 + 2 \tan\left(\frac{\pi x}{2}\right)$ for $-1 < x < 1$. 1

Explain why $g(3) = 0$.

(iii) Show that the gradient of the tangent to the curve $y = g(x)$ at the point $(3, 0)$ is $\frac{1}{\pi}$. 2

- (b) A particle is executing simple harmonic motion between points A and B .
 O is the centre of motion. Initially, the particle is at C , 10 units to the left of O .
At this time, it is moving to the right at 6 m/s and its acceleration is 0.625 m/s^2 .



The displacement of the particle after t seconds is given by $x = A \cos(nt + \alpha)$ for some constants A , n and α .

(i) Show that $n = \frac{1}{4}$. 1

(ii) Find the values of A and α . 3

(iii) M is the midpoint of OB . Find the time when the particle is first at M . 2

Question 14 continues on page 15

- (c) Car A and car B are travelling in the same direction along a straight, level road at constant speeds V_A and V_B respectively. Initially, car A is behind car B , but is travelling faster.

When car A is exactly D metres behind car B , car A applies its brakes, producing a constant acceleration $-k \text{ m/s}^2$, where $k > 0$.

- (i) Using calculus, show that the speed of car A after it has travelled D metres under braking is given by: 2

$$v = \sqrt{V_A^2 - 2kD}$$

- (ii) The distances travelled in t seconds by car A (after braking) and car B , are given by $x_A = V_A t - \frac{1}{2}kt^2$ and $x_B = V_B t$ respectively. 2
(Do NOT prove this).

Prove that the cars will collide if $V_A - V_B \geq \sqrt{2kD}$.

End of paper

2016 Extension 1 Mathematics

Trial HSC Solutions

Multiple Choice

Q1	D
Q2	D
Q3	A
Q4	D
Q5	C
Q6	C
Q7	C
Q8	D
Q9	C
Q10	B

Question 1:

$$\begin{aligned}\int \frac{1}{\sqrt{1-4x^2}} dx &= \int \frac{1}{\sqrt{4\left(\frac{1}{4}-x^2\right)}} dx \\&= \frac{1}{2} \int \frac{1}{\sqrt{\frac{1}{4}-x^2}} dx \\&= \frac{1}{2} \sin^{-1}(2x) + C\end{aligned}$$

Question 2:

$$\begin{aligned}x &= \frac{-5(9)+2(3)}{5-2} \\&= \frac{-39}{3} \\&= 13\end{aligned}$$

Question 3:

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{x \cos 3x}{\sin 2x} &= \lim_{x \rightarrow 0} \frac{x}{\sin 2x} \times \lim_{x \rightarrow 0} (\cos 3x) \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{2x}{\sin 2x} \times 1 \\
&= \frac{1}{2}
\end{aligned}$$

Question 4:

$$\begin{aligned}
x(x+5) &= 36 \\
x^2 + 5x - 36 &= 0 \\
(x+9)(x-4) &= 0 \\
x &= 4 \quad (\text{as } x > 0) \\
\therefore BD &= 4 \text{ cm}
\end{aligned}$$

Question 5:

$$\begin{aligned}
v^2 &= 4 + 4x - 2x^2 \\
&= 2(2 + 2x - x^2) \\
&= 2(3 - [x-1]^2)
\end{aligned}$$

This is in the form $v^2 = n^2(a^2 - [x-c]^2)$ where $a = \sqrt{3}$ and $c = 1$.

OR at the ends of the motion, $v = 0$:

$$\begin{aligned}
4 + 4x - 2x^2 &= 0 \\
x^2 - 2x - 2 &= 0 \\
(x-1)^2 &= 3 \\
x &= 1 \pm \sqrt{3}
\end{aligned}$$

So centre of motion is at $x = 1$ and amplitude is $\sqrt{3}$

Question 6:

$$1 + \sqrt{4 - x^2} > 0$$

$$\text{So } \sqrt{4 - x^2} > 0$$

$$\sqrt{(2-x)(2+x)} > 0$$

$$\text{As } 0 \leq \sqrt{4 - x^2} \leq 2$$

$$1 \leq 1 + \sqrt{4 - x^2} \leq 3$$

$$\ln(1) \leq \ln(1 + \sqrt{4 - x^2}) \leq \ln(3)$$

$$0 \leq y \leq \ln(3)$$

$$\text{Domain: } -2 \leq x \leq 2$$

$$\text{Range: } 0 \leq y \leq \ln(3)$$

Question 7:

$$\alpha\beta + \beta\gamma + \alpha\gamma = 2 \quad \text{and} \quad \alpha\beta\gamma = \frac{9}{2}$$

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{2}{\left(\frac{9}{2}\right)} \\ &= \frac{4}{9} \end{aligned}$$

Question 8:

$$\frac{dy}{dx} = -\frac{1}{x^2} \times -\frac{1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}}$$

$$\begin{aligned} u &= \frac{1}{x} \\ &= x^{-1} \end{aligned}$$

$$\frac{du}{dx} = -\frac{1}{x^2}$$

$$y = \cos^{-1} u$$

$$\frac{dy}{du} = -\frac{1}{\sqrt{1 - u^2}}$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{x^2} \times \frac{-1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \\ &= \frac{1}{x^2} \times \frac{\sqrt{x^2}}{\sqrt{x^2 - 1}} \\ &= \frac{1}{x^2} \times \frac{|x|}{\sqrt{x^2 - 1}} \\ &= \frac{1}{|x|\sqrt{x^2 - 1}} \end{aligned}$$

Question 9:

A. $LHS = \sin \frac{\pi}{2} \cos x + \cos \frac{\pi}{2} \sin x$ $= \cos x$ $RHS = \cos \frac{\pi}{2} \cos x + \sin \frac{\pi}{2} \sin x$ $= \sin x$ $\neq LHS$	B. $LHS = \sin x \cos \frac{3\pi}{2} + \cos x \sin \frac{3\pi}{2}$ $= -\cos x$ $RHS = \cos \pi \cos x + \sin \pi \sin x$ $= -\cos x$ $= LHS$
C. $LHS = \sin x \cos x$ $= \frac{1}{2} \sin 2x$ $-\frac{1}{2} \leq \frac{1}{2} \sin 2x \leq \frac{1}{2}$ $-\frac{1}{2} \leq \sin x \cos x \leq \frac{1}{2}$	D. $LHS = 2 + 2 \sin x - \cos^2 x$ $= 2 + 2 \sin x - (1 - \sin^2 x)$ $= \sin^2 x + 2 \sin x + 1$ $= (\sin x + 1)^2$ ≥ 0

Question 10:

$$\frac{dC}{dt} = 40 \text{ m/s}$$

$$C = 2\pi r$$

$$\frac{dC}{dr} = 2\pi$$

$$A = \pi r^2$$

$$\frac{dA}{dr} = 2\pi r$$

When $C = 100\pi$:

$$2\pi r = 100\pi$$

$$r = 50$$

$$\frac{dC}{dt} = \frac{dC}{dr} \times \frac{dr}{dt}$$

$$40 = 2\pi \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{20}{\pi} \text{ m/s}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 2\pi r \times \frac{20}{\pi}$$

$$= 40r \text{ m}^2 / \text{s}$$

When $r = 50$:

$$\frac{dA}{dt} = 40 \times 50$$

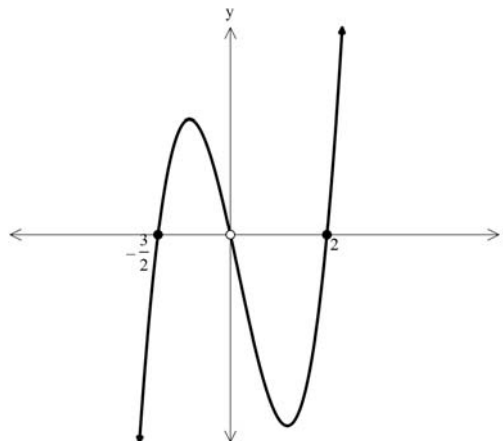
$$= 2000 \text{ m}^2 / \text{s}$$

Question 11:

(a) Solve $2x - 1 \geq \frac{6}{x}$.

3

$$\begin{aligned} x^2(2x-1) &\geq 6x \\ 2x^3 - x^2 - 6x &\geq 0 \\ x(2x^2 - x - 6) &\geq 0 \\ x(2x+3)(x-2) &\geq 0 \\ -\frac{3}{2} \leq x < 0, \quad x \geq 2 \end{aligned}$$



(b) (i) Show that $\sqrt{3} \sin x + \cos x$ can be written in the form $2 \cos\left(x - \frac{\pi}{3}\right)$.

3

$$\begin{aligned} \sqrt{3} \sin x + \cos x &\equiv A \cos(x - \alpha) \\ &\equiv A \cos x \cos \alpha + A \sin x \sin \alpha \end{aligned}$$

Equating coefficients of $\cos x$ and $\sin x$:

$$A \cos \alpha = 1$$

$$A \sin \alpha = \sqrt{3}$$

$$\begin{aligned} A &= \sqrt{1^2 + (\sqrt{3})^2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \tan \alpha &= \sqrt{3} \\ \alpha &= \frac{\pi}{3} \end{aligned}$$

$$\therefore \sqrt{3} \sin x + \cos x \equiv 2 \cos\left(x - \frac{\pi}{3}\right)$$

- (ii) Hence, or otherwise, find the value(s) of x for which $\sqrt{3} \sin x + \cos x$ is a minimum in the interval $0 \leq x \leq 2\pi$. 2

$$0 \leq x \leq 2\pi$$

$$-\frac{\pi}{3} \leq x - \frac{\pi}{3} \leq \frac{5\pi}{3}$$

Min. value of $2 \cos\left(x - \frac{\pi}{3}\right)$ is -2 .

$$2 \cos\left(x - \frac{\pi}{3}\right) = -2$$

$$\cos\left(x - \frac{\pi}{3}\right) = -1$$

$$x - \frac{\pi}{3} = \pi$$

$$x = \frac{4\pi}{3}$$

- (c) Use the substitution $u = \sqrt{x}$ to evaluate $\int_1^9 \frac{1}{x + \sqrt{x}} dx$. 3

$$u = \sqrt{x}$$

$$= x^{\frac{1}{2}}$$

Let $\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$

$$= \frac{1}{2\sqrt{x}}$$

$$2\sqrt{x} du = dx$$

$$dx = 2u du$$

$$\int_1^9 \frac{1}{x + \sqrt{x}} dx = \int_1^3 \frac{2u}{u^2 + u} du$$

$$= \int_1^3 \frac{2}{u+1} du$$

$$= 2[\ln(u+1)]_1^3$$

$$= 2(\ln 4 - \ln 2)$$

$$= 2 \ln 2$$

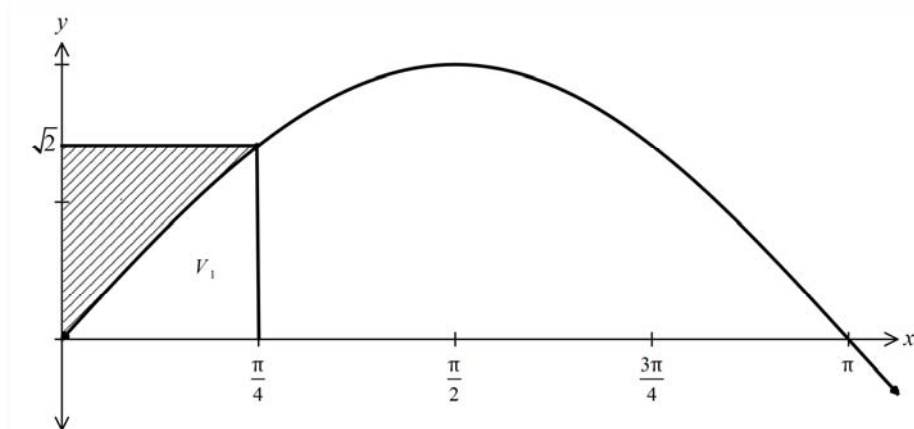
When $x = 1$, $u = 1$

When $x = 9$, $u = 3$

- (d) The diagram below shows the shaded region bounded by the curve $y = 2 \sin x$ and the y axis for $0 \leq y \leq \sqrt{2}$.

The region is rotated about the x axis to generate a solid of revolution.

- (i) Show that the volume, V , of the solid is given by $V = \frac{\pi^2}{2} - \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x dx$. 1



$$V_1 = \pi \int_0^{\frac{\pi}{4}} (2 \sin x)^2 dx$$

$$= \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x dx$$

$$\text{Shaded volume} = \pi \times (\sqrt{2})^2 \times \frac{\pi}{4} - \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x dx$$

$$V = \frac{\pi^2}{2} - \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x dx$$

- (ii) Find the exact value of V .

3

$$V = \frac{\pi^2}{2} - \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x dx$$

$$= \frac{\pi^2}{2} - \pi \int_0^{\frac{\pi}{4}} (2 - 2 \sin 2x) dx$$

$$= \frac{\pi^2}{2} - \pi [2x + \cos 2x]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi^2}{2} - \pi \left\{ \left(\frac{\pi}{4} + \cos \frac{\pi}{2} \right) - (0 + \cos 0) \right\}$$

$$= \frac{\pi^2}{2} - \pi \left\{ \frac{\pi}{4} - 1 \right\}$$

$$= \frac{\pi^2}{2} - \frac{\pi^2}{4} + \pi$$

$$= \pi \text{ units}^3$$

Using:

$$\cos 2x = 1 - 2 \sin^2 x$$

$$2 \sin^2 x = 1 - \cos 2x$$

$$4 \sin^2 x = 2 - 2 \cos 2x$$

Question 12:

(a) (i) Factorise $P(x) = 2x^3 + 3x^2 - 1$, given that $x = -1$ is a root.

2

$(x+1)$ is a factor of $P(x)$

$$2x^3 + 3x^2 - 1 = (x+1)(ax^2 + bx + c)$$

Equating coefficients:

$$a = 2 \text{ and } c = -1$$

$$b + a = 3$$

$$b + 2 = 3$$

$$b = 1$$

$$2x^3 + 3x^2 - 1 = (x+1)(2x^2 + x - 1)$$

$$= (x+1)(2x-1)(x+1)$$

$$= (x+1)^2(2x-1)$$

(ii) Solve the equation $\sqrt{x^2 + 2x + 5} = x + \sqrt{2x + 3}$.

3

$$\sqrt{x^2 + 2x + 5} = x + \sqrt{2x + 3}$$

Square both sides:

$$\cancel{x^2} + \cancel{2x} + 5 = \cancel{x^2} + 2x\sqrt{2x+3} + \cancel{2x} + 3$$

$$2 = 2x\sqrt{2x+3}$$

$$x\sqrt{2x+3} = 1$$

$$x^2(2x+3) = 1$$

$$2x^3 + 3x - 1 = 0$$

$$(x+1)^2(2x-1) = 0$$

Test $x = \frac{1}{2}$:

$$LHS = \sqrt{\left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right) + 5}$$

$$= \frac{5}{2}$$

$$RHS = \frac{1}{2} + \sqrt{2\left(\frac{1}{2}\right) + 3}$$

$$= \frac{5}{2}$$

$$= LHS$$

Test $x = -1$:

$$LHS = \sqrt{(-1)^2 + 2(-1) + 5}$$

$$= 2$$

$$RHS = -1 + \sqrt{2(-1) + 3}$$

$$= 0$$

$$\neq LHS$$

The only answer is $x = \frac{1}{2}$

- (b) Bobby has a can of cola at a temperature of 23°C . He places the can in a fridge which has a temperature of 3°C .

After t minutes, the temperature, c (in $^{\circ}\text{C}$), of the can of cola satisfies:

$$\frac{dc}{dt} = -\frac{1}{25}(c-3)$$

- (i) Show that $c = 3 + ae^{-\frac{t}{25}}$ satisfies this equation, where a is a constant.

1

$$\frac{dc}{dt} = -\frac{1}{25}(c-3)$$

$$\begin{aligned} LHS &= \frac{dc}{dt} \\ &= \frac{d}{dt} \left(3 + ae^{-\frac{t}{25}} \right) \\ &= -\frac{a}{25} e^{-\frac{t}{25}} \end{aligned}$$

$$\begin{aligned} RHS &= -\frac{1}{25}(c-3) \\ &= -\frac{1}{25} \left(3 + ae^{-\frac{t}{25}} - 3 \right) \\ &= -\frac{a}{25} e^{-\frac{t}{25}} \\ &= LHS \end{aligned}$$

So $c = 3 + ae^{-\frac{t}{25}}$ satisfies this equation.

- (ii) Bobby would like to drink the can of cola when its temperature is 5°C .

3

If he put the can in the fridge at 8:50 a.m, when should he drink it?

Give your answer to the nearest minute.

When $t = 0$:

$$\begin{aligned} 23 &= 3 + a(1) \\ a &= 20 \end{aligned}$$

When $c = 5$:

$$\begin{aligned} 5 &= 3 + 20e^{-\frac{t}{25}} \\ 2 &= 20e^{-\frac{t}{25}} \\ \frac{1}{10} &= e^{-\frac{t}{25}} \\ e^{\frac{t}{25}} &= 10 \\ \frac{t}{25} &= \ln(10) \\ t &= 25 \ln(10) \\ &= 57.5646 \dots \text{min} \end{aligned}$$

Bobby should drink the cola at 9:48 a.m.

(c) (i) Prove by mathematical induction that $\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$ for integers $n \geq 1$

3

Step 1: Prove true for $n = 1$

$$\begin{aligned} LHS &= 1^3 \\ &= 1 \end{aligned}$$

$$\begin{aligned} RHS &= \frac{1}{4} \times 1^2 (1+1)^2 \\ &= \frac{1}{4} \times 4 \\ &= 1 \\ &= LHS \end{aligned}$$

Proven true for $n = 1$

Step 2: Assume true for $n = k$.

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4}k^2(k+1)^2$$

Step 3: Prove true for $n = k + 1$

$$\text{RTP } 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$$

$$\begin{aligned} LHS &= 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 \\ &= \frac{1}{4}k^2(k+1)^2 + (k+1)^3 \text{ by assumption} \\ &= \frac{1}{4}(k+1)^2[k^2 + 4(k+1)] \\ &= \frac{1}{4}(k+1)^2[k^2 + 4k + 4] \\ &= \frac{1}{4}(k+1)^2(k+2)^2 \\ &= RHS \end{aligned}$$

By principle of mathematical induction, proven true for $n \geq 1$

(ii) Deduce that $2^3 + 4^3 + 6^3 + \dots + (2n)^3 = 2n^2(n+1)^2$.

1

$$\begin{aligned} 2^3 + 4^3 + 6^3 + \dots + (2n)^3 &= \sum_{r=1}^n (2r)^3 \\ &= 8 \sum_{r=1}^n r^3 \\ &= 8 \times \frac{1}{4}n^2(n+1)^2 \\ &= 2n^2(n+1)^2 \end{aligned}$$

(iii) Hence, or otherwise, find a simplified expression for:

2

$$1^3 + 3^3 + 5^3 + \dots + (2n-1)^3$$

$$\begin{aligned} 1^3 + 3^3 + 5^3 + \dots + (2n-1)^3 &= (1^3 + 2^3 + 3^3 + \dots + (2n-1)^3 + (2n)^3) - (2^3 + 4^3 + 6^3 + \dots + (2n-2)^3 + (2n)^3) \\ &= \sum_{r=1}^{2n} r^3 - \sum_{r=1}^n (2r)^3 \\ &= \frac{1}{4}(2n)^2(2n+1)^2 - 2n^2(n+1)^2 \\ &= n^2(2n+1)^2 - 2n^2(n+1)^2 \\ &= n^2[(2n+1)^2 - 2(n+1)^2] \\ &= n^2[4n^2 + 4n + 1 - 2n^2 - 4n - 2] \\ &= n^2(2n^2 - 1) \end{aligned}$$

Question 13:

- (a) The point $P(3p^2, 6p)$ lies on the parabola with $y^2 = 12x$ and the point S is the focus of this parabola. The point $Q(3q^2, 6q)$, where $p \neq q$, also lies on the parabola.

The tangent to the parabola at the point P and the tangent to the parabola at Q meet at the point R , as shown in the diagram below.

The equation of the tangent at point P is given by $y = \frac{1}{p}x + 3p$.

(Do NOT prove this).

- (i) Prove that $SP = 3(1 + p^2)$

1

For $y^2 = 12x$, S is $(3, 0)$

$$\begin{aligned} SP &= \sqrt{(3p^2 - 3)^2 + (6p)^2} \\ &= \sqrt{9p^4 - 18p^2 + 9 + 36p^2} \\ &= \sqrt{9p^4 + 18p^2 + 9} \\ &= \sqrt{9(p^4 + 2p^2 + 1)} \\ &= 3\sqrt{(p^2 + 1)^2} \\ &= 3(1 + p^2) \end{aligned}$$

Equation tangent at P: $y = \frac{1}{p}x + 3p$

Equation tangent at Q: $y = \frac{1}{q}x + 3q$

At R:

$$\frac{1}{p}x + 3p = \frac{1}{q}x + 3q$$

$$\frac{1}{p}x - \frac{1}{q}x = 3q - 3p$$

$$x \left(\frac{q-p}{pq} \right) = 3(q-p)$$

$$x = 3pq$$

When $x = 3pq$:

$$y = \frac{1}{p}(3pq) + 3p$$

$$= 3q + 3p$$

$$= 3(p+q)$$

Coordinates of R are $(3pq, 3(p+q))$

$$LHS = SR^2$$

$$= (3pq - 3)^2 + (3[p+q])^2$$

$$= 9p^2q^2 - 18pq + 9 + 9(p^2 + 2pq + q^2)$$

$$= 9p^2q^2 - \cancel{18pq} + 9 + 9p^2 + \cancel{18pq} + 9q^2$$

$$= 9(p^2q^2 + p^2 + q^2 + 1)$$

$$= 9(p^2[q^2 + 1] + 1[q^2 + 1])$$

$$= 9(p^2 + 1)(q^2 + 1)$$

$$RHS = SP \times SQ$$

$$= 3(1 + p^2) \times 3(1 + q^2)$$

$$= 9(p^2 + 1)(q^2 + 1)$$

$$= LHS$$

$$\therefore SR^2 = SP \times SQ$$

(b) In the diagram, EB is parallel to DC . Tangents from B meet the circle at A and C .

Let $\angle BCA = \alpha$

Prove that:

(i) $\angle BCA = \angle BFA$.

2

$\angle CDA = \angle BCA$ (angle between tangent and chord is equal to angle in the alternate segment)

$$= \alpha$$

$\angle BFA = \angle CDA$ (corresponding angles; $EB \parallel DC$)

$$= \alpha$$

$$\therefore \angle BCA = \angle BFA$$

(ii) $ABCF$ is a cyclic quadrilateral.

1

$\angle BCA = \angle BFA$ (proven)

$\therefore ABCF$ is a cyclic quadrilateral (angles in the same segment)

(c) Find a general solution of the following equation:

2

$$\sin\left(2x + \frac{\pi}{4}\right) = \frac{\sqrt{3}}{2}$$

$$\sin\left(2x + \frac{\pi}{4}\right) = \frac{\sqrt{3}}{2}$$

$$\left(2x + \frac{\pi}{4}\right) = n\pi + (-1)^n \frac{\pi}{3}$$

$$2x = n\pi + (-1)^n \frac{\pi}{3} - \frac{\pi}{4}$$

$$x = \frac{n\pi}{2} - \frac{\pi}{8} + (-1)^n \frac{\pi}{6}$$

$$= \frac{4n\pi - \pi}{8} + (-1)^n \frac{\pi}{6}$$

$$= \frac{\pi(4n-1)}{8} + (-1)^n \frac{\pi}{6}$$

- (d) A bowl is formed by rotating $y = x^{2n}$ (where n is an integer $n \geq 1$) about the y axis as shown.

The bowl is initially filled with water to a depth h . The bowl has a small hole in its bottom (initially plugged).

When the hole is unplugged, water flows at a rate described by:

$$\frac{dV}{dt} = -\pi a^2 \sqrt{2gh}$$

where a is the radius of the hole and g is a constant.

- (i) Show that the volume, V , of the water in the bowl is given by: 2

$$V = \frac{n\pi}{n+1} \cdot h^{\frac{1+n}{n}}$$

where h is the depth of the water in the bowl.

$$\begin{aligned} y &= x^{2n} \\ (x^{2n})^{\frac{1}{2n}} &= y^{\frac{1}{2n}} \\ x &= y^{\frac{1}{2n}} \\ V &= \pi \int_0^h x^2 dy \\ &= \pi \int_0^h \left(y^{\frac{1}{2n}} \right)^2 dy \\ &= \pi \int_0^h y^{\frac{1}{n}} dy \\ &= \pi \left[\frac{y^{\frac{1}{n}+1}}{\frac{1}{n}+1} \right]_0^h \\ &= \pi \left\{ \frac{h^{\frac{1}{n}+1}}{\frac{1}{n}+1} - 0 \right\} \\ &= \pi \left\{ \frac{h^{\frac{1+n}{n}}}{\frac{1+n}{n}} \right\} \\ &= \frac{n\pi}{1+n} \cdot h^{\frac{1+n}{n}} \end{aligned}$$

(ii) When the hole is unplugged, find the expression which describes the rate at which the water level is changing with respect to time.

2

$$\begin{aligned}
 V &= \frac{n\pi}{n+1} \cdot h^{\frac{1+n}{n}} \\
 &= \frac{n\pi}{n+1} \cdot h^{\frac{1}{n}+1} \\
 \frac{dV}{dh} &= \frac{n\pi}{n+1} \cdot \left(\frac{1}{n}+1\right) h^{\frac{1}{n}} \\
 &= \frac{n\pi}{n+1} \left(\frac{1+n}{n}\right) h^{\frac{1}{n}} \\
 &= \pi h^{\frac{1}{n}}
 \end{aligned}
 \quad \left| \quad
 \begin{aligned}
 \frac{dV}{dt} &= \frac{dV}{dh} \times \frac{dh}{dt} \\
 -\pi a^2 \sqrt{2gh} &= \pi h^{\frac{1}{n}} \times \frac{dh}{dt} \\
 \frac{dh}{dt} &= \frac{-a^2 \sqrt{2gh}}{h^{\frac{1}{n}}}
 \end{aligned}$$

(iii) If the water level falls at a constant rate with respect to time, find the value of n .

1

$$\begin{aligned}
 \frac{dh}{dt} &= \frac{-a^2 \sqrt{2gh}^{\frac{1}{2}}}{h^{\frac{1}{n}}} \\
 &= -a^2 \cdot \sqrt{2g} \cdot h^{\frac{1}{2} - \frac{1}{n}}
 \end{aligned}$$

For $\frac{dh}{dt}$ to be a constant:

$$\begin{aligned}
 \frac{1}{2} - \frac{1}{n} &= 0 \\
 n &= 2
 \end{aligned}$$

Question 14:

- (a) Consider a function $f(x)$. The domain of $y = f(x)$ is $a < x < b$ and the range is all real y . Assume that $f'(x)$ exists for $a < x < b$ and is non-zero.

Let $g(x)$ be the inverse function of $f(x)$. Assume that $g'(x)$ exists.

- (i) Show that $g'(x) = \frac{1}{f'(g(x))}$. 2

Using $f(g(x)) = x$

Differentiate both sides:

$$f'(g(x))g'(x) = 1$$

$$g'(x) = \frac{1}{f'(g(x))}$$

- (ii) Consider $f(x) = 3 + x^2 + 2 \tan\left(\frac{\pi x}{2}\right)$ for $-1 < x < 1$. 1

Explain why $g(3) = 0$.

$$f(0) = 3 + 0^2 + 2 \tan 0$$

$$= 3$$

Since $g(x)$ is the inverse of $f(x)$, $g(3) = 0$

- (iii) Show that the gradient of the tangent to the curve $y = g(x)$ at the point $(3, 0)$ is $\frac{1}{\pi}$. 2

$$g'(x) = \frac{1}{f'(g(x))}$$

$$g'(3) = \frac{1}{f'(g(3))}$$

$$= \frac{1}{f'(0)}$$

$$f(x) = 3 + x^2 + 2 \tan\left(\frac{\pi x}{2}\right)$$

$$f'(x) = 2x + 2 \cdot \frac{\pi}{2} \sec^2\left(\frac{\pi x}{2}\right)$$

$$= 2x + \pi \sec^2\left(\frac{\pi x}{2}\right)$$

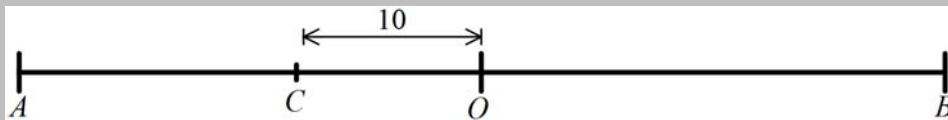
$$f'(0) = 2(0) + \pi \sec^2 0$$

$$= \pi$$

$$\text{So } g'(3) = \frac{1}{\pi}$$

and the gradient of the tangent at $x = 3$ is $\frac{1}{\pi}$.

- (b) A particle is executing simple harmonic motion between points A and B .
 O is the centre of motion. Initially, the particle is at C , 10 units to the left of O .
 At this time, it is moving to the right at 6 m/s and its acceleration is 0.625 m/s^2 .



The displacement of the particle after t seconds is given by $x = A \cos(nt + \alpha)$
 for some constants A , n and α .

- (i) Show that $n = \frac{1}{4}$.

1

$$x = A \cos(nt + \alpha)$$

$$\dot{x} = -An \sin(nt + \alpha)$$

$$\ddot{x} = -An^2 \cos(nt + \alpha)$$

$$= -n^2 \times A \cos(nt + \alpha)$$

$$= -n^2 x$$

When $x = -10$, $\ddot{x} = 0.625$:

$$0.625 = -n^2 \times -10$$

$$n^2 = 0.0625$$

$$n = 0.25$$

$$x = A \cos\left(\frac{1}{4}t + \alpha\right)$$

$$\dot{x} = -\frac{A}{4} \sin\left(\frac{1}{4}t + \alpha\right)$$

When $t = 0$:

$$-10 = A \cos(\alpha) \quad \text{--- (1)}$$

$$6 = -\frac{A}{4} \sin(\alpha)$$

$$-24 = A \sin(\alpha) \quad \text{--- (2)}$$

To find A :

$$A = \sqrt{(-10)^2 + (-24)^2}$$

$$= 26$$

As both $\sin \alpha$ and $\cos \alpha$ are negative, α is in 3rd quadrant.

To find α (sub into 1):

$$-10 = 13 \cos \alpha$$

$$\cos \alpha = -\frac{5}{13}$$

$$\alpha \approx \pi + \cos^{-1}\left(\frac{5}{13}\right)$$

$$\approx 4.3176$$

As the amplitude is 26, at M , $x = 13$:

$$13 = 26 \cos\left(\frac{1}{4}t + 4.3176\ldots\right)$$

$$\frac{1}{2} = \cos\left(\frac{1}{4}t + 4.3176\ldots\right)$$

$$\left(\frac{1}{4}t + 4.3176\ldots\right) = \frac{\pi}{3}, \frac{5\pi}{3}, \dots$$

$$\left(\frac{1}{4}t + 4.3176\ldots\right) = \frac{5\pi}{3} \quad (\text{as } t \geq 0)$$

$$t \approx 3.67\text{s}$$

- (c) Car A and car B are travelling in the same direction along a straight, level road at constant speeds V_A and V_B respectively. Initially, car A is behind car B, but is travelling faster.

When car A is exactly D metres behind car B, car A applies its brakes, producing a constant acceleration $-k \text{ m/s}^2$, where $k > 0$.

- (i) Using calculus, show that the speed of car A after it has travelled D metres under braking is given by: **2**

$$v = \sqrt{V_A^2 - 2kD}$$

$$\ddot{x} = -k$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -k$$

$$\frac{1}{2} v^2 = -kx + C$$

When $x = 0$, $v = v_A$:

$$\frac{1}{2} v_A^2 = C$$

$$\frac{1}{2} v^2 = -kx + \frac{1}{2} v_A^2$$

$$v^2 = -2kx + v_A^2$$

When $x = D$:

$$v^2 = -2kD + v_A^2$$

$$v = \sqrt{v_A^2 - 2kD} \quad (\text{since speed cannot be negative})$$

(ii) The distances travelled in t seconds by car A (after braking) and car B,

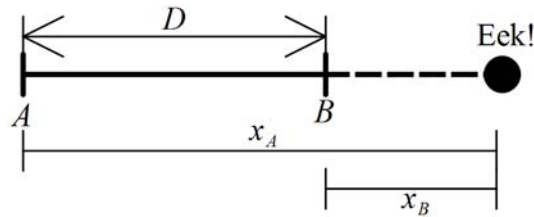
2

are given by $x_A = V_A t - \frac{1}{2}kt^2$ and $x_B = V_B t$ respectively.

(Do NOT prove this).

Prove that the cars will collide if $V_A - V_B \geq \sqrt{2kD}$.

When car A is exactly D m behind B, upon braking:



$$x_B = x_A - D:$$

$$V_B t = V_A t - \frac{1}{2}kt^2 - D$$

$$-\frac{1}{2}kt^2 + (V_A - V_B)t - D = 0$$

$$t = \frac{(V_B - V_A) \pm \sqrt{(V_A - V_B)^2 - 4 \cdot \left(-\frac{1}{2}k\right)(-D)}}{2 \left(-\frac{1}{2}k\right)}$$

$$= \frac{(V_B - V_A) \pm \sqrt{(V_A - V_B)^2 - 2kD}}{-k}$$

Real roots occur when $\Delta \geq 0$:

$$\Delta = (V_A - V_B)^2 - 2kD$$

$$(V_A - V_B)^2 - 2kD \geq 0$$

$$(V_A - V_B)^2 \geq 2kD$$

$$(V_A - V_B) \geq \sqrt{2kD} \quad (\text{as speed is positive})$$